

Cats

PHYLUM	Chordata
CLASS	Mammalia
ORDER	Carnivora
FAMILY	Felidae
SPECIES	37

With a lithe, muscular body, acute senses, highly evolved teeth and claws, lightning reflexes, and camouflage coloration, cats are model hunters. In fact, cats are the most specialized of the mammalian flesh eaters. They are unusual in that all species appear remarkably

similar: the differences between tigers and the domestic cat, for example, are surprisingly small. Cats are found throughout Eurasia, Africa, and the Americas (the domestic cat is found worldwide), from alpine heights to deserts. Many species live in forests. All except the largest cats are expert climbers, and several are excellent swimmers. Most cats are solitary.

Anatomy

Cats have a rounded face and a relatively short muzzle (but a wide gape). The large jaw muscles help deliver a powerful bite, and the long canines are used for stabbing and gripping. The carnassials, modified cheek teeth that slice bones and tendons, are highly developed. Cats are covered with soft fur, which is often striped or spotted, and have a tail that is haired, flexible, and usually long. There are 5 digits on the front feet and 4 on the back feet, and each digit has a curved, retractable claw for holding prey. The claws are normally retracted, which helps keep

them sharp. However, when required (during climbing, for example), they spring forward via a mechanism similar to a jackknife. The naked pads on the soles of the feet are surrounded by hair, which assists with silent stalking.

Senses

All cats have keen senses. Large, forward-facing eyes enable them to judge distances accurately. The pupils can contract to a slit or to a pinhole (depending on species) in bright light and can dilate widely for excellent night vision. The ears are large, mobile, and funnel-shaped

to draw in sounds made by prey, while long, stiff, highly sensitive whiskers aid navigation and night hunting. The sense of smell is also well developed, and in the roof of the mouth is a “smell-taste” organ, called the Jacobson’s organ, which detects sexual odors. Secretions from scent glands on the cheeks and forehead, under the tail, and between the claws communicate information such as age and sex.

Hunting techniques

While some cats actively search for prey, others conceal themselves and await passing victims. Many employ a combination of these 2 methods. In either case, the cat’s fur usually provides camouflage: a tiger’s stripes, for example, blend in with tall grasses, while many forest-living species are spotted, to mimic the effect of sunlight through leaves.

The distance a cat will chase its quarry varies between species. Heavier-built cats, such as tigers, prefer to stalk and pounce; the cheetah can use an explosive speed—up to 68 mph (110 kph) for short distances. Some small cats, such as the serval, hunt in long grass, and use “jack-in-the-box” leaps to surprise and flush out their prey.

CONSERVATION

The entire cat family is listed by the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), which regulates trade in live animals or their body parts. The main reason for this measure is habitat destruction. Land is converted to agricultural use, reducing and fragmenting the natural environment in which cats live. Many cat species require large areas to maintain an adequate food supply. Cats are also hunted for their fur, bones, and other remains for traditional oriental medicines. In a few species—such as the North American bobcat—the overall population is stable. However, despite intensive action to protect them, many other cats, including the tiger and the lion, are in sharp decline.

Cats hunt any animal they can catch and overpower. The big cats specialize in prey larger than themselves, and are capable of dragging a carcass some distance to a safe feeding spot. Smaller cats seek out rodents and birds—some, such as the fishing cat, wade into streams and scoop out fish.

TERRITORIAL CONFLICT

All cats are territorial and will fight if their scent marks and vocal warnings are ignored. These male jaguars are threatening one another, and a conflict will ensue if neither of them gives way. The flattened ears show fear, and dilated pupils and bared teeth indicate aggression.

